

Ryan Qin

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Education

University of Waterloo

Sept 2024 - Jun 2029

Bachelor of Applied Science in **Electrical Engineering**, Honours, Co-op

Technical Skills

Languages: C, C++, C#, Assembly, Verilog, Java, Python, HTML/CSS, JavaScript

Technologies: Robotics, RTOS, Linux, STM32, DSP, FSM, FPGA, TCP/IP, MongoDB, Node.js, React

Development Tools: TensorFlow, PyTorch, Android, AWS, Git, Cadence, Altium, MATLAB, DSO, Fusion 360

Professional Experiences

HSIO Compliance Engineer | TensorFlow, OpenCL, FPGA, C++, C#, Cadence, DSO, I²C
AMD

May 2026 - Present

- Designing **FPGA-based** USB 2.0 I/O testing board, aimed to **autonomously** test Link Layer and drop/droop compliance
- Developing **STM32-based I²C** module for **automatic** HDMI **EDID** setting, helping to reduce manual rewiring work
- Optimizing electrical Compliance analysis calculations using **OpenCL**, utilizing GPU parallelism for consecutive test runs
- Implementing **USB 4.0 test-automation** software in **C#**, designed to automate instruments over **TCP/IP** protocol

Electrical Engineer | LtSpice, Altium, EPLAN, C++, TCP/IP
ATS Corporation

Sep 2025 - Dec 2025

- Managed **full-cycle** design and verification for **four parallel** electronics projects, ensuring **on-time, spec-compliant** delivery
- Developed an **inductively isolated** pulse conditioner board in **Altium**, protecting PLC against stray voltage while **reducing 80% of Sub-GHz noise** from turbocharger's hall-effect tachometer
- Developed a **TCP/IP-enabled STM32** pneumatic leak-testing control board with a **thermal flow sensor** and **customized hardware drivers**, reducing power consumption by **40%** while decreasing unit's size by **50%**
- Designed and implemented **PID control loops** and **dither circuits** for a fuel nozzle testing system's pressure regulator, enabling **25%** more accurate simulation of turbojet engine operating conditions.

Electrical Coordinator | C#, .Net, MySQL, COMSOL
WSP

Jan 2025 - May 2025

- Recognized by **WSP Canada's** Rail and Transit **VP** for OCS engineering Software's **innovation intake** potential
- Independently developed Overhead Contact System (**OCS**) engineering software in **C#**, reducing Calculation time by **40%**
- Optimized traction power substation's cooling system layout in **COMSOL FEA**, aimed to extend transformer life by **10 years**

Projects

TakeOne  | CAN-BUS, Python, C++, TCP/IP

Sep 2026

- Developed an **AI-powered** film-taking robot, achieving **Finalist** at **Hack The North**
- Architected the Robots Communication and 42V/12V Power Delivery Network, ensuring stable and reliable drivetrain system
- Implemented AI-based IK robotics arm processing
- Designed Real-Time Control loop for robotic arm

TTL/FPGA Processor  | Verilog, Vivado, RTL, FSM, FPGA, DSP, RISC-V, TTL

Dec 2025 - Jan 2026

- Implemented **pipelined vector/tensor operations** in Verilog/SystemVerilog, exploiting **FPGA DSP** slices to increase compute throughput per clock cycle
- Designed and built a **custom CPU** from **discrete TTL logic**, implementing the ALU, register files, control unit, and clocking architecture from **first principles**
- Extended the processor architecture in **Verilog** for **FPGA deployment**, adding pipelining, forwarding, memory/port mapped I/O and SPI flash

Autodump  | I/O system Design, C++, TCP/IP, SSH, Python, OpenCV

Nov 2025

- Developed an **autonomous** self-emptying trash robot, achieving an **overall 3rd** place at **MakeCU**
- Engineered an Arduino-based serial **distributed I/O** system, focused on **simplifying** software-hardware integration and **reducing PWM-signal jitter and integrity** for servo and ultrasonic sensors

- Designed **ArUco** marker navigation system in **OpenCV**, allowing the robot to **autonomously navigate** back to home position

Competition RC 🏁 | PyTorch, FreeRTOS, LTSpice, AVR Assembly

May 2025

- Built an RC car with **50%** recycled material; selected as the **top qualified finalist** out of 15 teams
- Designed a **custom** airsoft motor controller using MOSFET stages, relays, and Arduino for PWM signal conversion, achieving a **fast lap record of 17 seconds** compared to **average 25 seconds** lap time
- Implemented a **lane-keeping algorithm**, combining **PID** and **PyTorch**-based vision to adjust steering from USB camera